THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS TWO-CHAPTER TEST SET A

Chapters 13 & 14: **Two Variables, One Line, Math of Space: Surface Area and Volume** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The graph of a linear equation in two variables is: 1

(a) a parabola **(b)** a circle **(c)** a single point **(d)** a straight line

2. The line ax + by + c = 0 always passes through the origin when: 1

(a) a = 0 **(b)** c = 0 **(c)** a = b **(d)** b = 0

3. **Assertion (A):** The slope of the line x = 3 is 0. 1

Reason (R): For the line x = 3 the run between any two points is 0, so the slope is undefined.

(a) Both A and R are true and R is the correct explanation of A

(b) Both A and R are true but R is not the correct explanation of A

(c) A is true but R is false

(d) A is false but R is true

4. The solution of the pair 2x + 3y = 5 and x − 2y = 6 is: 1

(a) (4, -1) **(b)** (-1, 4) **(c)** (-4, -1) **(d)** (4, 1)

5. If x and y satisfy 3x + 2y = 14 and x − y = -2, then the value of 2x + y is: 1

(a) 6 **(b)** 11 **(c)** 9 **(d)** 8

6. The total surface area of a cuboid of length 12 cm, width 5 cm and height 4 cm is: 1

(a) 240 cm^2 **(b)** 256 cm^2 **(c)** 128 cm^2 **(d)** 138 cm^2

7. The volume of a cuboid measuring 15 cm $\times$ 6 cm $\times$ 4 cm is: 1

(a) 380 cm^3 **(b)** 348 cm^3 **(c)** 25 cm^3 **(d)** 360 cm^3

8. Taking $\pi = \frac{22}{7}$, the curved surface area of a cylinder of radius 7 cm and height 10 cm is: 1

(a) 1540 cm^2 **(b)** 462 cm^2 **(c)** 220 cm^2 **(d)** 440 cm^2

9. Taking $\pi = \frac{22}{7}$, the volume of a cylinder of radius 14 cm and height 3 cm is: 1

(a) 1848 cm^3 **(b)** 3696 cm^3 **(c)** 264 cm^3 **(d)** 924 cm^3

10. Taking $\pi = \frac{22}{7}$, the volume of a cone of base radius 7 cm and height 24 cm is: 1

(a) 1232 cm^3 **(b)** 3696 cm^3 **(c)** 1254 cm^3 **(d)** 616 cm^3

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A taxi charges a fixed amount ₹x plus ₹y per km. A 6 km ride costs ₹110 and a 10 km ride costs ₹170. Write a pair of linear equations for this. 2

12. Without solving, say how many solutions $2x + 3y = 9$ and $4x + 6y = 18$ have. Give a reason. 2

13. A cone and a cylinder have the same base radius and the same height. Find the ratio of their volumes. 2

14. Write the total surface area of a cylinder closed at both ends, and explain each part. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Meera has ₹5 and ₹10 coins, 30 coins in all, with a total value of ₹220. How many coins of each kind does she have? 3

16. A ball of radius 7 cm is to be painted. Find the area to be painted and the cost at ₹0.25 per cm^2 ($\pi = \frac{22}{7}$). 3

17. Use $F = \frac{9}{5}C + 32$ to find 3
(i) the Celsius value corresponding to 212 °F,
(ii) the temperature at which Celsius and Fahrenheit readings are equal.

OR

Solve $11x + 9y = 73$ and $9x + 11y = 67$ without using the substitution method directly, by first adding and subtracting the equations. 3

18. The radii of two spheres are in the ratio $3 : 5$. Find the ratio of their 3
(i) surface areas,
(ii) volumes.

OR

A hemispherical bowl has an inner radius of 10.5 cm. Find the volume of liquid it can hold in litres ($\pi = \frac{22}{7}$). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A tank is filled so that the water level y (in cm) after x minutes follows the line $y = 4x$. 5
(i) What is the slope and what does it mean?
(ii) How high is the water after 6 minutes?
(iii) After how many minutes does it reach 50 cm?
(iv) Does the point $(8, 30)$ lie on this line? Justify.
(v) The tank's inlet is changed so that the level is $y = 4x + 10$. What does the 10 represent?

OR

Case study: The graph of the line l passes through $A(0, -2)$ and $B(4, 6)$. 5
(i) Find its slope.
(ii) Find the equation of l.
(iii) Where does l meet the x-axis?
(iv) A second line $x + y = 4$ is drawn on the same graph. Find the point where the two lines meet.

20. **Case study:** A company packs the same volume of product in either a cube of side 8 cm or a cuboid of 16 cm $\times$ 8 cm $\times$ 4 cm. 5
(i) Show that both boxes have the same volume.
(ii) Find the surface area of each.
(iii) Which needs less cardboard and by how much?
(iv) Cardboard costs ₹0.50 per cm^2; find the extra cost for 100 cuboid boxes.
(v) Name one situation where a larger surface area is useful.

OR

A cone, a sphere and a cylinder each have radius r; the cone and the cylinder have height $2r$ (the sphere's diameter). Show that their volumes are in the ratio $1 : 2 : 3$. Verify with $r = 3$ cm (leave answers in terms of π). 5

END OF QUESTION PAPER — **Total: 40 Marks**